

Chicago North Side Car Repair Directory — Project 1 Report

CS 284 / CS 430 · Fall 2026 · Project 1 Team: Saron, Eyael, Tutu · MySQL 8.4.5 · Snapshot collected 2026-09-30

1. User, scope, rules, sources, and counts

Intended user: a North Side Chicago driver who needs a specific repair (e.g. brakes or transmission) and wants to know which nearby shops advertise it, and where that claim came from. **Scope:** independent repair/maintenance shops holding an active City of Chicago *Motor Vehicle Services License* in four community areas (Lake View, Lincoln Park, Lincoln Square, North Center). We include a shop only if its identity and address appear in the license file *and* a published page names at least one specific service. Collision-only shops, car washes, and retailers were excluded.

1. **BR1 Identity:** a business is one licensed shop at one selected physical location. (`business_name`, `street_address`) is unique, and branches are never counted twice.
2. **BR2 Relationships:** every service belongs to exactly one category (1:N). Businesses and services are M:N, resolved by `business_service`.
3. **BR3 Required values:** every business and every offering must cite a source (FK, NOT NULL) and a raw-row reference. Name, address, and city are required. Website is optional (NULL = none recorded).
4. **BR4 No duplicates:** a business–service pair is stored once (composite PK), however many pages repeat it.

Source mix: the City of Chicago license dataset (government registry) supplies identity and location for all 16 shops. Offerings come from 14 provider websites plus two third-party directories (SimpleTire, EuroRepairHQ). Four more publishers (Consumers' Checkbook and three chamber/community directories) were used for corroboration: 21 publishers in total, 17 cited by rows. **Final counts:** 16 businesses, 7 services, 2 categories, 60 distinct offerings, and 4 corroborated businesses (T1 verifies these). Raw captures are in `data/raw/` (unchanged city CSV, the 83-row manual observation log, and saved HTML of every cited page). The rules, before/after examples, and exclusions are in `data/cleaning_note.md`.

2. Model and one important schema choice

Five tables (ER diagram on page 3; keys summarized in the README): `service_category` 1:N `service`; `business` M:N `service` through the associative table `business_service`; `source` 1:N to both `business` and `business_service`. Deleting a business cascades to its offerings. Categories, services, and sources are RESTRICTed while still referenced.

Key choice: provenance lives on the offering row. Each `business_service` row carries its own `source_id`, a `raw_row_ref` (e.g. OBS:OBS011), and the observed wording exactly as published (e.g. "Transmission Services"), next to the standardized `service_id`. This keeps the *observation* (what the page said) separate from our *classification* (which of 7 services it maps to). Q2 can then return a URL and raw row for every answer, and a mapping decision such as the E&J typo "Diagnostic Resting" → Vehicle Diagnostics can be audited. The PK (`business_id`, `service_id`) deliberately does *not* include `source_id`: a second citation of the same offering must not create a second offering (T3).

3. Normalization

Original relation (one row per observed offering, as first collected): R(`business_name`, `street_address`, `zip`, `neighborhood`, `website`, `license_row`, `service_name`, `category_name`, `observed_wording`, `obs_row`, `source_url`, `publisher`, `access_date`). **Candidate key:** {`business_name`, `street_address`, `service_name`}. **Functional dependencies:** FD1 `business_name`, `street_address` → `zip`, `neighborhood`, `website`, `license_row`; FD2 `service_name` → `category_name`; FD3 `source_url` → `publisher`, `access_date`; FD4 `business_name`, `street_address`, `service_name` → `observed_wording`, `obs_row`, `source_url`. FD1 and FD2 are partial dependencies on the key (a 2NF violation). FD3 is transitive through FD4 (a 3NF violation).

Decomposition to 3NF: FD1 → `business` (surrogate `business_id`, with the natural key kept UNIQUE); FD2 → `service` + `service_category`; FD3 → `source`; FD4 → `business_service`(`business_id`, `service_id`, `source_id`, `raw_row_ref`, `observed_wording`). In every table, each non-key attribute depends on the whole key and nothing but the key. **Anomaly removed (update):** in R, Fred's website was repeated on 7 rows (one per offering). Changing it meant 7 updates, and missing one would leave contradictory websites. Now it is stored once in `business`. (R also had a deletion anomaly: deleting Midtown's single offering row would have erased Midtown's address.) **Reconnection:** the foreign keys rejoin the facts losslessly. `business_service.business_id` = `business.business_id`, `...service_id` = `service.service_id`, `service.category_id` = `service_category.category_id`, and `...source_id` = `source.source_id` rebuild every original row (this is exactly the join in Q2).

4. Findings, limitation, and follow-up

Finding 1 (Q3, threshold 7): Vehicle Diagnostics is the most widely advertised service (14 of 16 shops), followed by Brake Repair (11), Tire Service (9), Oil Change (8), and A/C & Heating Repair (7). Transmission Repair (6) and Wheel

Alignment (5) fall below the threshold.

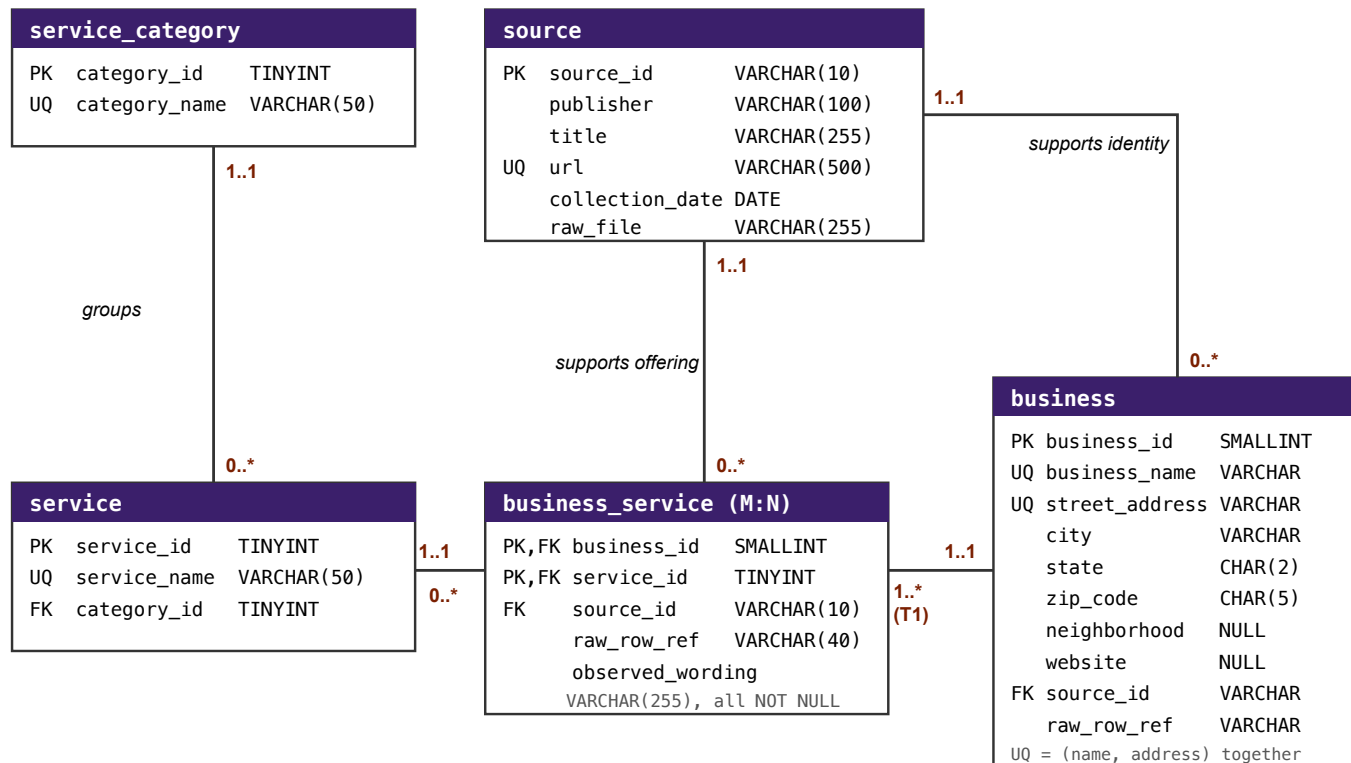
Finding 2 (Q4 with Wheel Alignment, Q2 with Transmission Repair): 11 of 16 shops have *no alignment offering recorded*, including both Lincoln Square shops (Mechanista, Speedline). In our sample, every recorded alignment offering is in Lake View, Lincoln Park, or North Center. Transmission work is recorded at 6 shops, each traceable to a source URL and raw row (e.g. Bucaro Brothers → SRC03 → OBS011).

Limitation: absence means *not recorded*, not "not offered". Evidence mostly comes from home pages, and a shop may do alignments without listing them (Cassidy lists only "Alignment Inspection", which we deliberately did not map). The sample is 16 shops from 20 checked candidates in four neighborhoods. Low provider counts therefore describe our sample, not unmet demand. **Follow-up question:** do the 11 shops without a recorded alignment offering actually perform alignments? This can be checked on their service pages or by phone, and then compared with the full Lincoln Square license list.

Contributions and AI use: see README (contribution table; AI-use statement: Claude/Claude Code assisted with source discovery, SQL, tests, and drafting, and every offering was verified against saved page snapshots).

ER diagram (implemented schema)

PK = primary key, FK = foreign key, UQ = unique. Labels at each line end give min..max participation. A mandatory side (min 1) comes from a NOT NULL FK. "1..* (T1)" = every business has at least one offering, enforced by the data rules and checked in test T1.



Reading the relationships

- service_category 1 : N service. Each service has exactly one category (FK NOT NULL); a category may have 0..* services.
- business M : N service, resolved by business_service. Each offering row has exactly one business and one service; a business has 1..* offerings (T1), and a service has 0..* offerings.
- source 1 : N business and source 1 : N business_service. Every row cites exactly one source (FK NOT NULL); a source may be cited 0..* times (SRC18–SRC21 are corroboration-only).
- ON DELETE: business → business_service CASCADE; all other FKs RESTRICT. PK(business_id, service_id) blocks duplicates.